

OPIPE-05: Business & Security Event Pipelines

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Processing Business Transactions and Security Events at Ingestion

Business events and security events are two OpenPipeline scopes with distinct governance requirements. Business events drive revenue analytics, conversion tracking, and SLO calculations. Security events drive compliance reporting, threat detection, and audit trails. Both benefit from ingestion-time processing — enrichment, routing, and metric extraction — but the rules are different.

This notebook covers the Business Events and Security Events scopes in OpenPipeline. For log processing, see **OPLOGS-01: OpenPipeline Fundamentals**. For compliance and masking patterns, see **OPLOGS-08: Security & Data Protection** and **OPMIG-08: Security & Masking**.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail and business events enabled
Permissions	storage:bizevents:read , storage:events:read , openpipeline:configurations:write
Data	Business events ingested via OneAgent RUM, API, or OpenTelemetry
Recommended	OPIPE-01 (multi-scope architecture), BIZEV series for business event analytics

1. Business Events Scope

Business events represent **user transactions and business actions** — purchases, sign-ups, page views, API calls, form submissions. They are ingested through:

- **OneAgent RUM** — Automatic capture of user actions on web and mobile apps
- **Business Events API** — Custom events sent from backend services
- **OpenTelemetry** — Span-derived business events

Key Business Event Fields

Field	Description	Example
event.type	The business event type	com.myapp.purchase.completed
event.provider	The source system	www.myshop.com
event.category	Event classification	purchase , signup , pageview
event.group	Logical grouping	checkout-flow , onboarding
Custom attributes	Business-specific data	order.total , product.category , user.tier

Discovering Your Business Event Landscape

```
// Business event volume by type and provider
fetch bizevents, from:-24h
| summarize event_count = count(), by:{event.type, event.provider}
| sort event_count desc
| limit 20
```

```
// Business event trend by type over 24h
fetch bizevents, from:-24h
| makeTimeseries event_count = count(), by:{event.type}, interval:1h
```

2. Business Event Enrichment

Business events often arrive with technical identifiers that need business context. OpenPipeline enrichment adds meaningful fields at ingestion.

Common Enrichment Patterns

Condition	Enrichment Field	Value	Purpose
<code>event.type</code> contains <code>"purchase"</code>	<code>business.domain</code>	<code>"commerce"</code>	Domain classification
<code>event.provider</code> == <code>"mobile-app"</code>	<code>channel</code>	<code>"mobile"</code>	Channel attribution
<code>event.provider</code> == <code>"www.myshop.com"</code>	<code>channel</code>	<code>"web"</code>	Channel attribution
Custom: <code>order.total</code> > 1000	<code>order.tier</code>	<code>"high_value"</code>	Revenue tier classification
Custom: <code>user.country</code> matches <code>{"US", "CA"}</code>	<code>region</code>	<code>"north_america"</code>	Geographic grouping

Security Context for Business Events

Sensitive business data (revenue figures, user details) should be access-controlled:

Condition	<code>dt.security_context</code> Value	Rationale
<code>event.type</code> contains <code>"payment"</code>	<code>"finance-team"</code>	Payment data restricted to finance
<code>event.type</code> contains <code>"signup"</code>	<code>"marketing-team"</code>	User acquisition data for marketing
Default	<code>"product-team"</code>	General product analytics

3. Business Event Metric Extraction

Extract metrics from business events for dashboards, SLOs, and alerting.

Example: E-Commerce Metrics

Metric Key	Aggregation	Condition	Dimensions
bizevent.purchase_count	count	event.type == "purchase.completed"	event.provider , channel
bizevent.purchase_revenue	sum(order.total)	event.type == "purchase.completed"	event.provider , channel
bizevent.cart_abandonment	count	event.type == "cart.abandoned"	event.provider
bizevent.signup_count	count	event.type == "user.signup"	channel , region

Conversion Funnel Metrics

Track progression through a multi-step funnel by extracting a count metric at each stage:

Funnel Stage	Event Type	Metric Key
Product viewed	product.viewed	funnel.stage_1_views
Added to cart	cart.item_added	funnel.stage_2_adds
Checkout started	checkout.started	funnel.stage_3_checkout
Purchase completed	purchase.completed	funnel.stage_4_purchase

```
// Conversion funnel from business events (last 24h)
fetch bizevents, from:-24h
| summarize views = countIf(event.type == "product.viewed"),
  adds = countIf(event.type == "cart.item_added"),
  checkouts = countIf(event.type == "checkout.started"),
  purchases = countIf(event.type == "purchase.completed")
```

4. Security Events Scope

Security events include threat detections, vulnerability findings, audit records, and compliance-relevant actions. They require special handling:

- **Never drop** — Security events must be retained for compliance, even if volume is high
- **Never sample** — Every security event is potentially significant
- **Long retention** — Compliance frameworks (SOC 2, HIPAA, PCI-DSS) require 1-7 year retention
- **Strict access control** — Only security teams should access security event data

Key Security Event Types

Event Type	Source	Description
Runtime vulnerability detections	OneAgent	Third-party library vulnerabilities
Attack detections	Application Security	SQL injection, command injection attempts
Audit events	Platform	Configuration changes, user actions
Custom security events	API ingestion	SIEM, IDS/IPS, firewall events

```
// Security event overview
fetch events, from:-24h
| filter event.kind == "SECURITY_EVENT" OR matchesValue(dt.system.bucket, "*security*")
| summarize event_count = count(), by:{event.type}
| sort event_count desc
```

5. Security Event Routing and Compliance

Security events should always be routed to **dedicated buckets** with extended retention and restricted access.

Routing Strategy

Pipeline	Routing Condition	Target Bucket	Retention	Security Context
vulnerability-detections	event.type matches <code>"*VULNERABILITY*"</code>	security_vulnerabilities	365 days	"security-team"
attack-detections	event.type matches <code>"*ATTACK*"</code>	security_attacks	365 days	"security-team"
audit-events	event.type matches <code>"*AUDIT*"</code>	security_audit	365 days	"security-team"
compliance-events	Custom compliance markers	compliance_events	2555 days (7 years)	"compliance-team"

Compliance Framework Requirements

Framework	Retention Requirement	Key Event Types
SOC 2	1 year minimum	Access logs, configuration changes, incident records
HIPAA	6 years	Access to PHI, authentication events, audit trails
PCI-DSS	1 year (3 months readily available)	Cardholder data access, authentication, network events
GDPR	As long as necessary	Data access, consent, deletion requests

6. Event-to-Metric Extraction for KPIs

Extract operational KPIs from both business and security events for dashboards and alerting.

Business KPIs

KPI	Metric Source	Extraction Rule
Conversion rate	<code>purchase.completed</code> / <code>product.viewed</code>	Extract count for each, compute ratio in dashboard
Average order value	<code>order.total</code> from <code>purchase.completed</code>	Extract avg aggregation
Revenue per hour	<code>order.total</code> from <code>purchase.completed</code>	Extract sum aggregation, interval 1h

Security KPIs

KPI	Metric Source	Extraction Rule
Attack attempts per hour	Attack detection events	Extract count, interval 1h
Open vulnerabilities	Vulnerability events	Extract countDistinct by CVE ID
Failed auth attempts	Authentication events	Extract count where <code>status == "FAILED"</code>

```
// Security event trend over 7 days
fetch events, from:-7d
| filter event.kind == "SECURITY_EVENT" OR matchesValue(dt.system.bucket, "*security*")
| makeTimeseries event_count = count(), by:{event.type}, interval:24h
```

Summary

In this notebook you learned:

- **Business events scope** — Sources, key fields, and discovery queries for transaction data
 - **Business event enrichment** — Adding business context, channel attribution, and security context at ingestion
 - **Metric extraction from business events** — Conversion funnels, revenue metrics, and operational KPIs
 - **Security events scope** — Never drop, never sample, always retain with extended retention
 - **Compliance-driven routing** — Dedicated buckets with retention aligned to SOC 2, HIPAA, PCI-DSS requirements
 - **Event-to-metric extraction** — KPI dashboards for both business and security monitoring
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Next Steps

Continue to **OPIPE-06: Cross-Scope Design Patterns** to learn how to correlate data across scopes, cascade processing from spans to events to metrics, and design a production-ready OpenPipeline architecture.

References

- [Business Events](#)
 - [OpenPipeline Business Events](#)
 - [Application Security](#)
 - [Data Retention Management](#)
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